Coding & Solution

Date	15 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Kavitha-2108/Smart-Lender-ML-Project
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, XGBoost, Pandas, Scikit-learn, Joblib

Key Features Implemented	• User Registration and Login • Loan Application Form • Loan Approval Prediction using Machine Learning • Input Validation • Prediction Result Display • Responsive Web Interface
Pending / Incomplete Features	• Loan History Dashboard • Email Notifications (Optional) • Advanced Analytics
Setup / Run Instructions	1. Install Python 3.x. 2. Install dependencies using pip install -r requirements.txt. 3. Run the application using python app.py. 4. Open http://127.0.0.1:5000 in your browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- The application is developed using **Python (Flask)** for the backend.
- The frontend is built with **HTML, CSS, and JavaScript** for an interactive user experience.
- **JavaScript** is used for client-side form validation and dynamic user interactions.
- **XGBoost** is used as the machine learning model for loan prediction.
- The project follows a modular coding structure and is maintained using Git and GitHub.